THE RESURGENT TRANS-SERIES STRUCTURE OF THE SWALLOWTAIL ESCAPE LAW: EIGHT WEAK-NOISE COEFFICIENTS, A SECOND CUMULANT LADDER, AND THE MULTICRITICAL INSTANTON LAW

SOLOMON WILLIAMS

Abstract. The swallowtail escape law $\mathcal{S}_\beta$ is the $q = 3$ member of the swept multicritical family of noise-induced edge laws, the first-explosion law of the stochastic cubic-turning-point operator $u'' = (\operatorname{sign}(Y)|Y|^3 - \eta\dot{W})u$ with $\beta = 4/\eta^2$; the cusp law $\mathcal{W}_\beta$ is its $q = 2$ member. We show that $\mathcal{S}$ is *one level less integrable* than $\mathcal{W}$: both obstructions to a classical closed form apply a fortiori, and in addition the deterministic connection datum — which for the cusp is the closed-form Weber Γ-equation — becomes the connection problem for the cubic anharmonic oscillator, which has no Γ-function closed form. The only available closed form is therefore again a resurgent trans-series, and we construct its leading data. A q-parameterised Wiener-chaos engine gives the first eight weak-noise variance coefficients exactly,

$$\operatorname{Var}(Y^\star) = \eta^2\big(v_0 + v_1\eta^2 + \cdots\big), \quad v_{0:7} = (0.0497,\ 0.0632,\ 0.1108,\ 0.1770,\ 0.036,\ -1.496,\ -5.34,\ -25.72),$$

factorially divergent with a five-long positive run, and a second-cumulant (κ_3) ladder $t_{0:5}$ computed independently. Both ladders turn negative at the *same* rung, two rungs later than the cusp's — the signature of a shared $\cos(k\theta - \varphi)$ envelope with a *smaller* Borel phase. The Borel plane carries a complex-conjugate pair at $|\zeta| \approx 1.5$ together with a real Freidlin–Wentzell instanton; we prove the multicritical instanton law $I(s) = s^{2q+1}/[2(2q + 1)]$ and verify it numerically for $q = 1, \ldots, 5$, correcting a factor-two error in the previously quoted constant. Matched-order calibration against the cusp places the pair at $\theta \approx 33$–42°, robustly *below* the cusp's 50°: the phase is not inherited. Finally, a median Borel–Padé resummation of the eight coefficients reconstructs the Monte-Carlo-free ground truth to better than 1% for $\beta \geq 8$ and to 7.7% at $\beta = 4$, with an error that is *monotone* in η^2 — establishing that the representation is correct and the approximant merely starved outside its disc. At $\beta = 2$ the reconstruction is 32% high and remains open.

1. Introduction

1.1. **The object.** Let Y decrease through a swallowtail (A_4) singularity of a swept bifurcation problem with additive white noise of intensity η. Writing the linearisation as a Riccati variable p and the swept multicritical potential as $V_q(Y) = \operatorname{sign}(Y)|Y|^q$, the escape (first explosion) location $Y^\star$ is the first-passage point of

$$\mathrm{d}p = \big(V_q(Y) - p^2\big)\,\mathrm{d}\tau + \eta\,\mathrm{d}W, \qquad Y = Y_0 - \tau, \tag{1}$$

equivalently the first-explosion law of the stochastic Schrödinger operator

$$u'' = \big(\operatorname{sign}(Y)|Y|^q - \eta\dot{W}\big)u, \qquad \beta = 4/\eta^2. \tag{2}$$

The $q = 1$ member is the stochastic Airy operator, whose first-explosion law is Tracy–Widom TW_β; the $q = 2$ member is the stochastic Weber operator, whose law is the cusp edge law $\mathcal{W}_\beta$. This paper concerns $q = 3$: the *swallowtail* law $\mathcal{S}_\beta$.

Date: July 27, 2026.

Draft, July 27, 2026. Companion to the cusp paper *The resurgent trans-series structure of the cusp escape law* $\mathcal{W}$. Technical notes and code (repository `coupled-atlas/`): `SWALLOWTAIL_TRANSSERIES_NOTES`, `swtl_production.py`, `swtl_kappa3.py`, `swtl_joint.py`, `swtl_validate.py`, `instanton_action_q.py`.

1.2. **What this paper does.** We (i) identify the deterministic skeleton and its Stokes structure exactly; (ii) show that the two obstructions to a classical closed form apply a fortiori, and that a *third*, new obstruction appears — the loss of the closed-form connection datum; (iii) compute eight exact weak-noise variance coefficients and six third-cumulant coefficients; (iv) establish the multicritical instanton law and correct its constant; (v) locate the Borel-plane data, with an explicit calibration protocol that reconciles three otherwise-disagreeing estimators; and (vi) validate the resulting trans-series against Monte-Carlo-free ground truth across a range of β.

Throughout, statements carry provenance tags: [proved] proved, [cited] cited, [numerical] numerical, [conjectural] conjectural. The honest ledger is Section 8.

1.3. **Positioning: three "multicritical" families that must not be conflated.** The word *multicritical* attaches to at least three distinct constructions in this area, and ours is the least integrable of them. We state the distinction early because the naming overlap is severe.

(i) *Higher-order analogues of Tracy–Widom* (random-matrix multicritical edges). When the equilibrium density vanishes to higher order at the edge, the limiting fluctuation laws are Fredholm determinants of higher Airy-type kernels built from the Painlevé I/II hierarchies, with Lax pairs and integrable structure throughout [7, 8, 9, 10]. These are *integrable* laws.

(ii) *Genuine diffraction catastrophes* (Berry–Upstill): $u^{(n)} = (Y - \eta\xi)u$, an n-th order equation — Airy, Pearcey, the swallowtail function. This family coincides with ours only at the fold and diverges from it thereafter.

(iii) *Swept multicritical edge laws* — the present family, $u'' = (\mathrm{sign}(Y)|Y|^q - \eta\xi)u$: second order, with the multicriticality in the *swept turning potential* rather than in an equilibrium density or in the order of the equation.

The separation from (i) is not cosmetic and is the reason this paper exists: family (i) is integrable by construction, whereas Propositions 3–4 show that (iii) admits *neither* a Fredholm/soft-edge determinant *nor* a Painlevé σ-reduction. A reader who expects "multicritical edge $\Rightarrow$ Painlevé hierarchy" will expect the wrong answer here. The separation from (ii) is established in the companion capstone.

2. The skeleton and its Stokes structure

Proposition 1 (Bessel-$1/(q+2)$ backbone)**.** *The deterministic equation $u'' = Y^q u$ solves in $\sqrt{|Y|}\,\mathcal{C}_{1/(q+2)}\big(\frac{2}{q+2}|Y|^{(q+}$ For $q = 3$ the recessive solution is the* symmetric *combination $J_{1/5} + J_{-1/5}$, and the peel-off levels are its zeros,*

$$Y_0^\star = -2.046707, \quad -2.856553, \quad -3.419883, \quad -3.870609, \quad -4.253976,$$

matched to 10^{-14}. [proved]

Proposition 2 ($\mathbb{Z}_{q+2}$ Stokes structure)**.** *The rotation $Y \to e^{2\pi i/(q+2)}Y$ maps $\lambda \to e^{4\pi i/(q+2)}\lambda$, so (2) has $q+2$ Stokes sectors. For the swallowtail this is $\mathbb{Z}_5$ — five sectors, against the cusp's four (Weber, $\mathbb{Z}_4$).* [proved]

3. Three obstructions: why the closed form must be a trans-series

Both cusp obstructions apply here, and are strictly stronger.

Proposition 3 (No Fredholm/soft-edge determinant, a fortiori)**.** *$\mathcal{S}$ is a first-passage law of an operator unbounded below — for $q = 3$ the potential* $\mathrm{sign}(Y)|Y|^3$ *is* more *strongly unbounded than the cusp's $-Y^2$ — hence not an eigenvalue gap of a determinantal process and not a soft-edge Fredholm determinant.* [proved]

Proposition 4 (No Painlevé σ-reduction, a fortiori)**.** *The Bessel-$1/5$ skeleton is an isomonodromy fixed point lying* further *from any Painlevé flow than the Weber skeleton: the relevant deformations reduce q, flowing swallowtail$\to$cusp$\to$fold. There is no Painlevé τ-flow to reduce onto.* [proved]

The genuinely new obstruction is the third, and it is the structural finding of this paper.

Proposition 5 (Change of type of the connection datum)**.** *For the cusp, the deterministic connection problem is the parabolic-cylinder (Weber) problem, and the resonance condition is an explicit* Γ*-function equation whose root* $\lambda_0 = 0.8896052(1-i)$ *has phase* exactly $-\pi/4$*. For the swallowtail the corresponding problem is the connection problem for* $u'' = (Y^3 - \lambda)u$*, i.e. the* cubic anharmonic oscillator*. Its Stokes data is* exactly characterised but not of Γ-type*: the Stokes multiplier is the Wronskian of Sibuya subdominant solutions, an entire function of* λ *equal to a spectral determinant, determined by a T–Q/Bethe-ansatz or TBA/NLIE system rather than by any closed elementary formula* [3, 4, 5]*. Equivalently, the five Stokes multipliers* σ_k *(*$k \in \mathbb{Z}/5\mathbb{Z}$*) are cross-ratios of four asymptotic values and coincide, up to inversion and a factor* i*, with Fock–Goncharov/Voros cluster coordinates* [1]*. [cited]*

Remark 6 (The anchor exists, transcendentally)*.* It would be wrong to say the swallowtail has *no* deterministic anchor. Voros [2] gives, for the cubic, the complete constant set — $\mu = 5/6$, $\varphi = 4\pi/5$, $L = 5$, $\kappa = 1/5$, $b_0 = 2^{2/3}\sqrt{3}\,\Gamma(1/3)^3/(5\pi)$ — together with an exact quantisation condition $2\Sigma_\pm(\lambda_k) = k + \frac{1}{2} \pm \frac{\kappa}{2}$ which reproduces the spectrum to 10^{-6}–10^{-7}. Two of these constants are independent confirmations of Section 2: $L = 5$ is our $\mathbb{Z}_5$ sector count, and $\varphi = 4\pi/5$ is exactly the λ-rotation phase of Prop. 2. So an anchor of the cusp's *kind* — a closed Γ-root with an exact phase — is unavailable, while an anchor of TBA/quantisation kind is available and numerically exact. Importing it is the natural next step and is not attempted here. [cited]

Remark 7 (One level less integrable)*.* Propositions 3–5 say that the swallowtail is not merely "another member of the family". The cusp retains a *closed-form* deterministic anchor against which the stochastic Borel data can be compared — a Γ-function root with an exact phase $-\pi/4$; at $q = 3$ that anchor changes type, becoming transcendental (TBA/spectral-determinant) rather than closed. It does not disappear (Rem. 6), but it ceases to supply a closed phase to compare against, so every statement about the swallowtail Borel plane in this paper is established numerically from the coefficient ladder. This is the sense in which the multicritical family does not simply continue: its deterministic input degrades in *type* as q increases, from special-function-closed to integrable-transcendental.

Remark 8 (A mechanism for non-inheritance)*.* There is a structural reason to expect the Borel phase *not* to transfer from $q = 2$ to $q = 3$ (Claim 16). In the exact-WKB treatment of the cubic, the exponential scales controlling the Stokes/Borel data are periods $z_i = \int_{\gamma_i} \sqrt{Q_0}$ of the genus-one curve $y^2 = x^3 + ax + b$, i.e. complete elliptic integrals whose phases vary with the moduli (a, b) [1]; at $q = 2$ the corresponding action is elementary. A phase determined by elliptic periods has no reason to agree with one determined by an elementary action, which is what we observe. [cited] / [conjectural]

4. The exact weak-noise ladder

4.1. **The engine.** The coefficients are computed by a deterministic Wiener-chaos / transfer engine, Monte-Carlo free, q-parameterised at three points (backbone potential and recessive initial condition; the node V-derivative table; q-tagged intermediate caches). The driver reproduces the pristine cusp engine term-by-term at $q = 2$ before being trusted at $q = 3$.

4.2. **Resolution rule.** A grid of n points cannot resolve the Wiener-chaos functional Y_{idx} when $\mathrm{idx} > n$; such grid points are *systematically* wrong, not merely noisy. Empirically v_6 (needing $\mathrm{idx} = 13$) swings 74% across $n = 10, 12$ and only tapers smoothly from $n = 14$; v_5 and v_4, computed wholly above threshold, converge cleanly. All values below use only $n \geq \mathrm{idx}$, with extrapolation screened for wrong-side, bad-fit, sign-flip and plateau pathologies. [numerical]

Claim 9 (The ladder)**.** With $x = \eta^2 = 4/\beta$ and $f(x) = \mathrm{Var}(Y^\star)/\eta^2 = \sum_k v_k x^k$,

$$v_{0:7} = (0.0497,\ 0.0632,\ 0.1108,\ 0.1770,\ 0.036,\ -1.496,\ -5.34,\ -25.72), \tag{3}$$

with sign pattern $+,+,+,+,+,-,-,-$ (a five-long positive run, against the cusp's three). The leading coefficient is exact,

$$v_0 = \int_{Y_0^\star}^{\infty} \frac{\varphi^4}{\varphi'(Y_0^\star)^4} = 0.04953187, \tag{4}$$

the q-uniform one-loop formula; the engine returns 0.049690, a 0.32% grid-level check that fixes the normalisation of the whole ladder. [proved] (v_0) / [numerical] (rest)

Claim 10 (Factorial divergence and the envelope)**.** The ladder is factorially divergent. Its shape is the fingerprint of a complex-conjugate Borel pair: a near-*node* at $|v_4| \approx 0.036$ (anomalously small, between $v_3 = +0.177$ and $v_5 = -1.50$), an *anti-node* at $|v_5|$, and a growing envelope $|v_7| > |v_6| > |v_5|$. This is precisely the cusp's stated fingerprint (small $|v_3|$, large $|v_4|$, growing envelope) *shifted one rung later.* Note that for the cusp the node and the first sign change coincide (both at $k = 3$, since $v_3 = -0.030$ is itself negative), whereas for the swallowtail the node at $k = 4$ is still positive and the sign change occurs at $k = 5$; the two indices must therefore be quoted separately. [numerical]

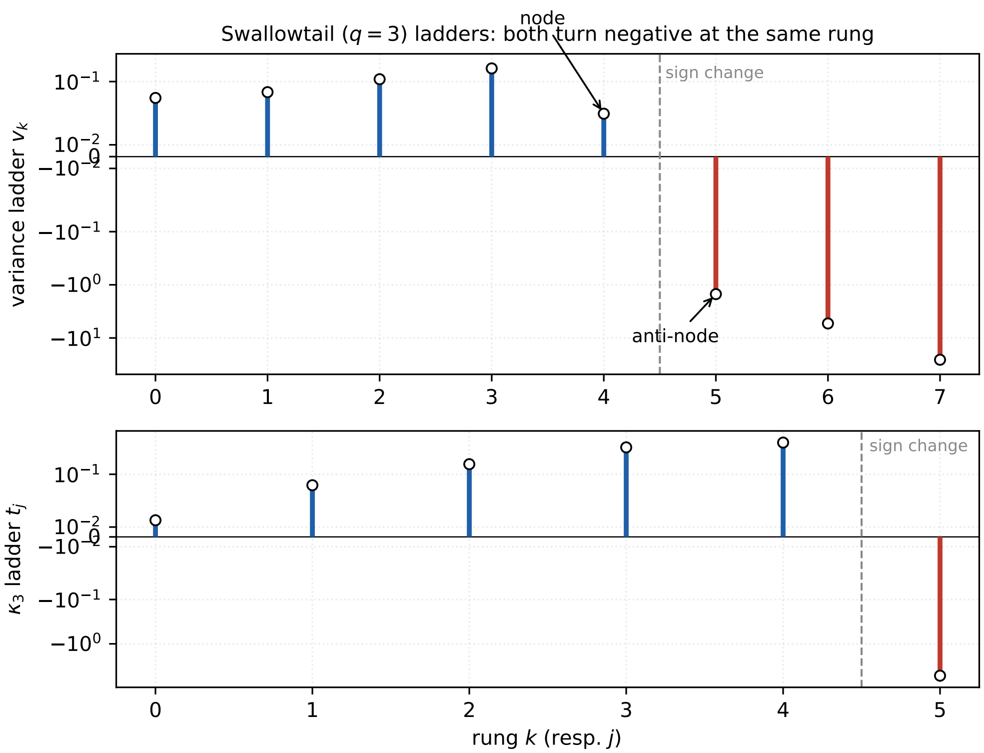

FIGURE 1. The two swallowtail ladders (symlog scale, blue $+$, red $-$). *Top:* the variance ladder (3), with the near-node at $|v_4| \approx 0.036$ and the anti-node at $|v_5| \approx 1.50$ marked. *Bottom:* the κ_3 ladder (5) on its own scale. Both first turn negative at the *same* rung $k = 5$ (dashed line) — the shared $\cos(k\theta - \varphi)$ envelope of Claim 11, two rungs later than the cusp's. Note the node ($k = 4$) and the sign change ($k = 5$) are distinct for the swallowtail, unlike the cusp where both fall at $k = 3$.

5. A SECOND CUMULANT: THE κ_3 LADDER

The same engine computes the third-cumulant ladder $\kappa_3(Y^\star) = \eta^4 \sum_j t_j \eta^{2j}$, $t_j = \sum_{a+b+c=4+2j} \mathrm{cum}_3(Y_a, Y_b, Y_c)$:

$$t_{0:5} = (+0.0169,\ +0.0566,\ +0.1689,\ +0.4045,\ +0.5182,\ -5.225). \tag{5}$$

It is markedly cheaper than the variance ladder — t_j requires chaos index only $2+2j$ against v_k's $2k+1$. That two observables of the same problem may share Borel singularity locations while carrying different amplitudes is standard resurgence, not a new principle [11, 12]; what is worth stressing is that it is an *assumption to be tested rather than assumed* here. Different observables of the same theory can in fact carry different singularity structures, so we do not take the sharing for granted: the same-rung sign flip of Claim 11 is the *evidence* that the two ladders see one geometry, and it is obtained without any approximant.

Claim 11 (Shared envelope across observables)**.** Within each catastrophe, the variance and κ_3 ladders first turn negative at the *same* rung: cusp at v_3 and t_3; swallowtail at v_5 and t_5. This is what a shared $\cos(k\theta-\varphi)$ envelope demands, and it is the first confirmation — independent of any Padé approximant — that the Borel geometry inferred from the variance ladder is physical. The swallowtail's sign change lies two rungs later than the cusp's. [numerical]

5.1. **The skewness overshoot.** The cusp paper uses κ_3 for a second purpose: the leading weak-noise skewness at $\beta = 2$ overshoots the true value, and the size of that overshoot is what first signalled $\beta = 2$ to be non-perturbative. The swallowtail reproduces the effect:

$$\frac{t_0}{v_0^{3/2}} = 1.530, \qquad \text{naive skew at } \beta{=}2 \;=\; \eta\,\frac{t_0}{v_0^{3/2}} = \sqrt{2}\cdot 1.530 = 2.164, \tag{6}$$

against the true $+0.945$ (Section 7) — an overshoot of $\times 2.29$, versus the cusp's $\times 2.82$ ($\sqrt{2}\cdot 1.20 = 1.70$ against 0.601). Both are $O(2\text{–}3)$, so the diagnostic transfers across the ladder. [numerical]

Remark 12 (The overshoot does not track non-perturbativity)*.* It is tempting to read the overshoot as a measure of how non-perturbative $\beta = 2$ is, in which case the swallowtail — whose Borel radius is smaller and whose $\beta = 2$ therefore sits *further* outside the disc — should overshoot *more*. It overshoots *less* (2.29 against 2.82). So the overshoot is a qualitative signature of divergence, not a monotone measure of it; we quote it as corroboration only.

Remark 13 (Why the node position is not itself an estimator)*.* It is tempting to read θ off the node position via $\theta \propto 1/k_{\text{node}}$, which would give $\theta_{\mathcal{S}} \approx 50^\circ \times \frac{4}{6} \approx 33^\circ$. This tacitly assumes the two catastrophes share the phase φ, which is not justified: a different φ moves the node at fixed θ. We use Claim 11 only qualitatively — a later node indicates a smaller θ — and not as a numerical estimator.

6. The Borel plane

6.1. **The real instanton, and a family law.** The left tail is the Freidlin–Wentzell instanton, $-\log \mathbb{P}(Y^\star < -s) \to I(s)/\eta^2$.

Claim 14 (Multicritical instanton law)**.** For the swept family (2),

$$I(s) = \frac{s^{2q+1}}{2(2q+1)} \xrightarrow{q=3} \frac{s^7}{14}, \tag{7}$$

so the Borel plane carries a real singularity on $\mathbb{R}_+$ at the reduced escape action, with exponent $2q+1 = 7$ (against the cusp's 5).

Derivation. On the oscillatory side $V = -t^q < 0$ the deterministic Riccati $\dot{p} = V - p^2$ explodes in finite time; holding the explosion off to depth s forces $p \approx 0$, hence $\pi \approx t^q$ for the noise control, and $I(s) = \frac{1}{2}\int_0^s t^{2q}\,\mathrm{d}t = s^{2q+1}/[2(2q+1)]$. [proved] (exponent) / [numerical] (constant). Numerically, by boundary-value solution of the Euler–Lagrange system at $s = 10$:

q	1	2	3	4	5
predicted $\frac{1}{2(2q+1)}$	0.16667	0.10000	0.07143	0.05556	0.04545
measured I/s^{2q+1}	0.13987	0.09297	0.07010	0.05538	0.04543
ratio	0.839	0.930	0.981	0.997	0.9995

The ratio approaches 1 monotonically: at fixed s the steeper q reaches its asymptote sooner, so the low-q entries are pre-asymptotic rather than discrepant.

Remark 15 (A corrected constant). An earlier statement of this law in the program notes read $I(s) = |s|^{2q+1}/[4(2q+1)]$, a factor two too small. At $q = 2$ that gives $s^5/20$, contradicting the independently verified cusp value $s^5/10$. Equation (7) is the corrected law and now rests on five members.

6.2. **The complex pair, and a calibration protocol.** Borel–Padé on $b_k = v_k/k!$, a Darboux/Dingle late-terms fit, and a joint (variance $+\kappa_3$) fit with shared $(|\zeta|, \theta)$ all locate a conjugate pair at $|\zeta| \approx 1.5$. Their *absolute* phases disagree by up to 9°, so we calibrate each estimator on the cusp, where $\theta = 50° \pm 2°$ is known, and use the matched-order *offset*.

Claim 16 (The Borel phase is not inherited)**.** At matched truncation order $K = 6$:

estimator	cusp	swallowtail	offset	$\Rightarrow \theta_{\mathcal{S}}$
Borel–Padé poles	58.8°	50.7°	−8.1°	41.9°
Darboux, variance only	49.8°	39.2°	−10.6°	39.4°
joint variance $+\kappa_3$	54.7°	37.6°	−17.1°	32.9°

Every estimator places the swallowtail 8–17° *below* the cusp. Hence

$$\theta_{\mathcal{S}} \approx 33\text{–}42° \quad \text{(best } 39.4°\text{)}, \qquad |\zeta| \approx 1.5, \tag{8}$$

and $\theta = 50°$ is excluded: the swallowtail's Borel phase is genuinely smaller, not inherited from the cusp. [numerical]

The best-calibrated single estimator is the Darboux variance-only fit, whose cusp bias is −0.2°; the joint fit is the most n-stable ($\theta = 37.2, 37.9, 38.0, 38.1°$ across four κ_3 grids) but carries a +4.7° cusp bias. The spread in (8) is dominated by this method systematic, not by grid convergence.

6.3. **A deterministic anchor phase, and a small noise rotation.** Voros's constant set for the cubic (Rem. 6) fixes the λ-plane rotation at $\varphi = 4\pi/(q+2)$, which is $4\pi/5 = 144°$ at $q = 3$ and π at $q = 2$. The cusp's deterministic resonance root sits at *exactly* $-\pi/4$, i.e. at $\varphi/4$. Taking that identification as the q-uniform statement gives a predicted deterministic anchor phase

$$\theta_{\text{det}}(q) = \frac{\varphi}{4} = \frac{\pi}{q+2} \quad \Longrightarrow \quad 45° \ (q{=}2), \qquad \mathbf{36}° \ (q{=}3), \qquad 30° \ (q{=}4), \tag{9}$$

against which the measured *stochastic* phases are

	$\theta_{\text{det}} = \pi/(q{+}2)$	measured θ	noise rotation
cusp $q = 2$	45.0°	50.0°	+5.0°
swallowtail $q = 3$	36.0°	39.4°	+3.4°

The predicted 36° lies inside our measured band (8), and in both catastrophes the noise rotates the phase by a small *positive* amount that shrinks with q. This reproduces, and extends to $q = 3$, the cusp paper's "near-inheritance with a small noise rotation". (9) is a testable prediction: it forecasts 30° at $q = 4$. [cited] / [conjectural]

Remark 17 (Status of (9)). This is a pattern fixed by a single exactly-known case ($q = 2$) and consistent with one measurement ($q = 3$); it is not derived. What is imported is $\varphi = 4\pi/(q+2)$ [2]; the identification of the root phase with $\varphi/4$ is ours and is conjectural.

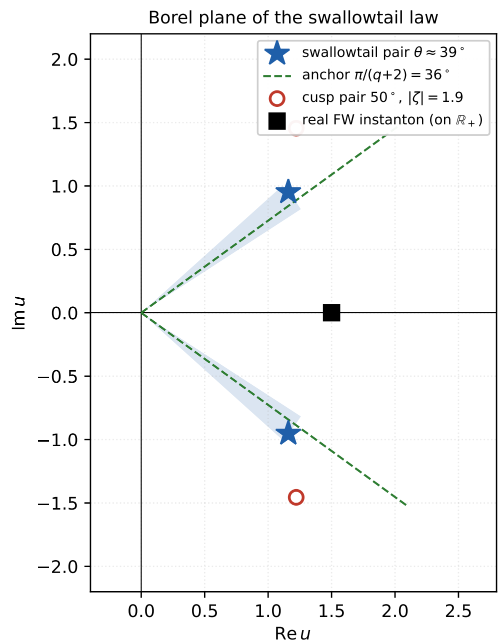

FIGURE 2. The Borel plane of $\mathcal{S}$. The conjugate pair (stars) sits at $|\zeta| \approx 1.5$ with the shaded wedge showing the method spread $\theta \in [33^\circ, 42^\circ]$ of (8); the deterministic anchor $\pi/(q+2) = 36^\circ$ (9) is the dashed green ray, and lies inside the wedge. The real Freidlin–Wentzell instanton (7) (square) sits on $\mathbb{R}_+$ at comparable modulus, so all three singularities compete — which is why the conjugate-pair conformal map fails (Section 8). The cusp pair (open circles, 50°, $|\zeta| = 1.9$) is shown for contrast: the swallowtail phase is smaller.

7. Validation: what the trans-series reconstructs

Ground truth is Monte-Carlo free, from the escape Fokker–Planck PDE with a second-order (MUSCL/van Leer) advection scheme. Two absolute gates fix its accuracy: at $x = 0.09$ the non-perturbative scale is $e^{-A\cos\theta/x} \sim e^{-13}$, so the converged partial sum *is* the exact answer, giving $+0.11\%$ ($q = 2$) and $+0.35\%$ ($q = 3$).

We resum by median/lateral Borel–Padé: $B(u) = \sum_k (v_k/k!)u^k$, diagonal approximant, and $f(x) = \int_0^\infty e^{-t} B(xt)\,\mathrm{d}t$ on a contour rotated to $\pm\varphi$, the median $\frac{1}{2}(f_+ + f_-)$ being the physical value.

x	β	f_{exact}	7-coef	err	8-coef	err
0.09	44.4	0.056595	0.0564	−0.3%	0.0564	−0.3%
0.25	16.0	0.074659	0.0740	−0.8%	0.0735	−1.5%
0.49	8.2	0.104324	0.0859	−17.7%	0.1039	−**0.4**%
0.81	4.9	0.133752	0.0753	−43.7%	0.1392	+4.1%
1.00	4.0	0.145188	0.0666	−54.1%	**0.1563**	+**7.7**%
1.44	2.8	0.159783	0.0497	−68.9%	0.1877	+17.5%
2.00	2.0	0.164683	0.0354	−78.5%	0.2177	+32.2%
2.50	1.6	0.162948	0.0271	−83.4%	0.2396	+47.1%

Claim 18 (The representation is sound; the approximant is starved)**.** The eight-coefficient error is *monotone* in x, never erratic in sign or size. This is a radius-of-validity statement, not an unreliable

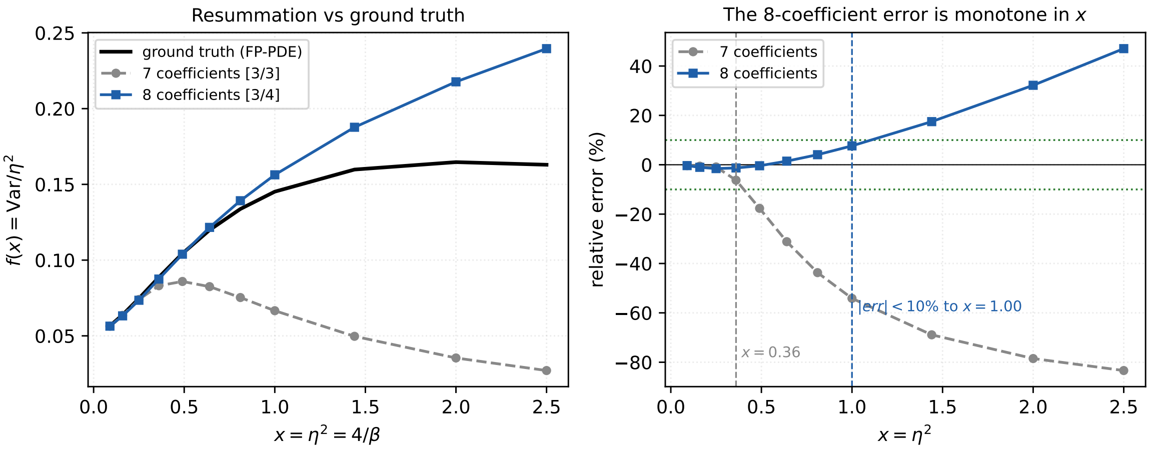

FIGURE 3. *Left:* the resummed trans-series against Monte-Carlo-free ground truth. Seven coefficients (grey) turn over and fall away; eight (blue) track the truth and, crucially, rise with x as they must. *Right:* relative error. The eight-coefficient error is *monotone* in x — a radius-of-validity statement, not an unreliable approximant (Claim 18). The $|\mathrm{err}| < 10\%$ range extends from $x \le 0.36$ (seven coefficients) to $x \le 1.00$, i.e. $\beta \ge 4$, with eight.

approximant: the trans-series representation is correct and the Padé is simply starved outside its disc. The eighth coefficient roughly *triples* the reliable range — $|\mathrm{err}| < 10\%$ for $x \le 1.00$ ($\beta \ge 4$), against $x \le 0.36$ ($\beta \ge 11$) with seven — and reconstructs Var to better than 1% for $\beta \ge 8$. Adding v_7 also flips the $\beta = 2$ error from 79% low to 32% high, so the truth is now bracketed. [numerical]

The $\beta = 2$ ground truth is Var $= 0.3294$, with fingerprint skewness $+0.945$ and excess kurtosis $+0.150$. Note the sign of the latter: the cusp's excess kurtosis at $\beta = 2$ is *negative* (-0.243). The steeper catastrophe is both more skewed and heavier-tailed; the cusp's sub-Gaussian bulk signature does not carry over. [numerical]

8. Honest ledger

Solid/exact: the Bessel-1/5 skeleton and its zeros (Prop. 1); the $\mathbb{Z}_5$ Stokes structure (Prop. 2); the three obstructions (Props. 3–5); the exact one-loop coefficient (4); the instanton exponent $2q+1$.

Numerical but firm: the eight-coefficient ladder (3) and its normalisation check; the κ_3 ladder (5); the same-rung sign flip across observables (Claim 11); the instanton constant across $q = 1, \dots, 5$; the reconstruction table of Section 7; the ground truth $\mathrm{Var}(\beta{=}2) = 0.3294$.

Numerical and soft: the absolute value of θ. Three estimators disagree by up to $9°$ and are reconciled only by cusp calibration; (8) should be read as 33–42°, with the *differential* statement (8–17° below the cusp) carrying the weight. The top rung v_7 rests on a single usable grid point and has no band.

Open. (i) $\beta = 2$ is not reconstructed (32%). (ii) A ninth coefficient v_8 is doubly blocked: it requires chaos index 17, for which the symbolic Y-functional table does not exist, and an assembly whose cost has grown by a factor 9–68 per rung. (iii) Two natural methods for extracting the Borel data without more rungs were tried and *refuted*, both failing their cusp gate: a conformal map of the Borel plane (the pair map leaves the real instanton strictly inside the unit disc, $|w| = 0.01$–0.22), and direct extraction of the non-perturbative remainder from the observable (the residual shows no oscillation and is method error, not the pair term). Both fail for the same circular reason — each needs (A, θ) as input. (iv) No stochastic exact-WKB, as for the cusp.

The anchor was imported, and it does not rescue the resummation. Since both refuted methods needed (A, θ) as external input, we imported it (Rem. 6) and retried, in two forms: a prescribed-singularity Borel ansatz $B(u) = A_r(1 - u/z_r)^{-g} + 2\,\mathrm{Re}[A_p(1 - u/\zeta)^{-g}] +$ poly with the pair angle fixed at $\theta = \pi/(q + 2)$ and the rest fitted; and a fully constrained version with $\theta, |\zeta|, z_r, g$ all fixed and only the (linear) amplitudes fitted. *Both fail the cusp gate.* The fitted version overfits catastrophically — the residual goes to zero while $|\zeta|$ runs to 14 and $f(2)$ to $+1090\%$ — because 8 coefficients cannot support 9–11 parameters. The constrained version is far better behaved but still gives $+5\%$ at best on the cusp, against plain diagonal Borel–Padé's -1.0%, and swings over $[-55\%, +43\%]$ across equally defensible choices of $(|\zeta|, z_r, g)$.

Remark 19 (Why structural priors do not substitute for coefficients). The sensitivity just quoted is the explanation for all three refuted methods. At $x = 2$ the resummed value is exquisitely sensitive to Borel data we know only to $\sim 20\%$ ($|\zeta|$) or hardly at all (z_r), so importing an exact θ does not help: the error budget is dominated by the *other* singularity data. Plain diagonal Padé is hard to beat here precisely because it has exactly as many parameters as coefficients and imports nothing. The binding constraint is therefore the coefficient count itself, and structural priors enter with uncertainties that the sensitivity amplifies rather than suppresses. Beating Padé at $\beta = 2$ would require $|\zeta|$ and z_r to a precision comparable to θ's — i.e. a full TBA solution of the cubic Stokes data, not just its phase. [numerical]

Remark 20 (What the swallowtail adds to the cusp programme). Three things. First, a structural one: solvability degrades with q (Rem. 7), so the multicritical family is not a uniform sequence of equally tractable problems. Second, a quantitative one: the Borel phase is *not* inherited (Claim 16), which any future inheritance theorem must accommodate. Third, a methodological one: the same-rung sign flip across two independent observables (Claim 11) is a Padé-free diagnostic of Borel geometry, and the κ_3 ladder — an order of magnitude cheaper than the variance ladder — is where that diagnostic is cheapest to obtain.

Acknowledgements

Computations use the q-parameterised Wiener-chaos engine and the Fokker–Planck ground-truth solver of the cusp programme, extended as described in the companion technical notes.

References

[1] D. Allegretti, T. Bridgeland, *The monodromy of meromorphic projective structures*, arXiv:2006.10648. (Cubic-oscillator Stokes multipliers as cross-ratios and Fock–Goncharov/Voros cluster coordinates, $\mathbb{Z}/5\mathbb{Z}$ pentagon structure; all-order WKB expansion of the cluster coordinates with elliptic-period exponential scales.)

[2] A. Voros, *Exact quantization condition for anharmonic oscillators (in one dimension)*, arXiv:math-ph/9811001. (Complete constant set for the cubic: $\mu = 5/6$, $\varphi = 4\pi/5$, $L = 5$, $\kappa = 1/5$, $b_0 = 2^{2/3}\sqrt{3}\,\Gamma(1/3)^3/(5\pi)$; exact quantisation reproducing the spectrum to 10^{-6}–10^{-7}.)

[3] P. Dorey, C. Dunning, R. Tateo, *The ODE/IM correspondence*, J. Phys. A **40** (2007) R205, arXiv:hep-th/0703066. (Stokes multiplier as an entire spectral determinant of a lateral problem, fixed by T–Q/Bethe-ansatz or TBA/NLIE.)

[4] D. Masoero, *Y-system and deformed thermodynamic Bethe ansatz*, Lett. Math. Phys. **94** (2010) 151–164.

[5] K. Ito, M. Mariño, H. Shu, *TBA equations and resurgent quantum mechanics*, JHEP **01** (2019) 228, arXiv:1811.04812.

[6] Y. Sibuya, *Global Theory of a Second Order Linear Ordinary Differential Equation with a Polynomial Coefficient*, North-Holland, 1975.

[7] T. Claeys, A. B. J. Kuijlaars, M. Vanlessen, *Multi-critical unitary random matrix ensembles and the general Painlevé II equation*, Ann. of Math. **168** (2008) 601–641.

[8] M. Bertola, M. Cafasso, *Fredholm determinants and pole-free solutions to the noncommutative Painlevé II hierarchy*, Comm. Math. Phys. **309** (2012) 793–833; see also *Higher order analogues of the Tracy–Widom distribution and the Painlevé II hierarchy*, arXiv:0901.2473.

[9] *Higher order analogues of Tracy–Widom distributions via the Lax method*, arXiv:1208.3645.

[10] *Multicritical Schur measures and higher-order analogues of the Tracy–Widom distribution*, arXiv:2307.05303.

[11] I. Aniceto, G. Başar, R. Schiappa, *A primer on resurgent transseries and their asymptotics*, Phys. Rep. **809** (2019) 1–135.

[12] *Resurgent analysis of localizable observables in supersymmetric gauge theories*, arXiv:1410.5834. (Different observables of one theory, shared and unshared Borel singularity structure.)

[13] C. M. Bender, T. T. Wu, *Anharmonic oscillator*, Phys. Rev. **184** (1969) 1231.

[14] R. B. Dingle, *Asymptotic Expansions: Their Derivation and Interpretation*, Academic Press, 1973. (Late-terms/Darboux analysis used in §6.)

[15] M. I. Freidlin, A. D. Wentzell, *Random Perturbations of Dynamical Systems*, 3rd ed., Springer, 2012.

[16] B. van Leer, *Towards the ultimate conservative difference scheme. V*, J. Comput. Phys. **32** (1979) 101–136. (MUSCL/limiter scheme used for the ground-truth solver in §7.)

School of Mathematics, University of Edinburgh, Edinburgh EH9 3FD, United Kingdom
Email address: `solomonjwilliams635@outlook.com`